



Reduction of Charge-Transfer Resistance via Artificial SEI Formation Using Electropolymerization of Borylated Thiophene Monomer on Graphite Anodes

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Modification of electrode surface using borylated-thiophene monomer resulted in the formation of a pre-formed solid electrolyte interface (SEI). The obtained ready-to use anodes did not show any loss in capacity during the first few cycles of charging and discharging. Herein, we report the synthesis and utilization of boron containing monomer to form an artificial SEI to reduce the interfacial resistance at SEI. Electropolymerization was performed via two methods, i) pre-modification and ii) in-situ modification. After the modification of the anodes, modified electrodes showed increased capacities, reduced capacity fading effect, and reduced interfacial resistance. Charge-discharge characteristics of half-cell with borylated thiophene polymer over graphite was studied and compared with graphite without any modification and polythiophene coated graphite.

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In lithium ion batteries, highly graphitized carbonaceous materials have been emphasized for studying the interfacial chemistry between the electrodes and electrolytes.¹⁻⁴ These graphitized carbon materials result in a high irreversible capacity loss after the first cycle. Several researchers have studied this aspect of the lithium ion battery i.e., reduction of irreversible capacity loss and improvement of the overall performance of the battery.⁵ During the first charge of the batteries, reduction of electrolyte leads to the formation of the Solid Electrolyte Interface (SEI) layer which enables effective intercalation/deintercalation into/from the electrodes. Ideally, SEI layer should be thin, conductive and stable in nature.^{6,7} However, generally used, Li and Li-C electrodes are passivated by the surface films in optimized electrolyte solutions. Therefore, less efficient charge-transfer between electrodes and electrolytes are inescapable.⁸ Generally, this irreversible decomposition of electrolyte solvent leads to reduction of electrochemical activity of lithium and Li-C anodes.^{8,9} These irreversible reactions are parasitic in nature^{10,11} and has to be inhibited in order to obtain a high performance battery.

Several approaches are examined to improve the battery performance such as designing novel electrolyte materials,^{12,13} development of high capacity electrode materials such as silicon^{14,15} or carbon-silicon composites.¹⁶⁻¹⁸ Among all this, surface modification of anodes is one of the practical approach to reduce the irreversible capacities arising because of an unstable SEI formation.¹⁹ Methods such as oxidation, fluorination, ceramic coating and electropolymerization have already been applied and studied. Among all these, electropolymerization of a conducting polymer with heterocyclic aromatic monomer units results in high thermal and chemical stability of interface. Also, the process of electropolymerization can be easily controlled and result into surface films with strong adhesion.²⁰

Use of heterocyclic aromatic monomer such as thiophene has gained significant attention and is studied well in this field. Its structural versatility and environmental stability of its doped and undoped states lead to their application in the field of electrodes, conductors and semiconductors.²¹ Modification of the electrodes using pyrrole,¹⁹ thiophene,¹⁹ derivatives of thiophene such as α -terthiophene,²² α -tetrathiophene²³ etc., has been reported. However, boron incorporation into thiophene or making borylated-thiophene compounds that can be electropolymerized to form a surface film over electrodes has not been studied till now. When boron is doped into the carbon lattice,²⁴ it results into oxidation resistant materials.²⁵ Incorporation of boron into monomers like thiophene can help in reduction of the electrode-electrolyte interfacial resistance. Boron being Lewis acidic in nature,

leads to generation of hole-carriers when doped onto graphite, thereby increasing its electrical conductivity and hence, discharge capacity.²⁶ In a similar manner, formation of a conducting polymer surface film doped with boron is expected to increase the overall electrical conductivity of graphite-polymer film electrode. This will result into a highly conductive SEI layer that will further enhance the lithiation and improve the performance of the battery.

To address the problem of poor cyclability and reversibility arising due to anodes, aim of this article is to modify the electrode surface with an electronically conductive polymer which will help in reducing the charge-transfer resistance at the electrode-electrolyte interface and lead to higher capacities. In the present work, the detailed synthesis of a low molecular weight boron monomer and its electropolymerization onto the graphite electrodes is studied. Electropolymerization has been carried out via two different methods,

- Pre-modification of electrodes via cyclic voltammetry and then, testing for battery applications,
- Direct in-situ electropolymerization via charge-discharge analysis. In-situ electropolymerization requires addition of the monomer as an additive with the electrolyte in the fabricated cell for charge-discharge.

This method of in-situ modification is advantageous as compared to pre-modified anodes as in this method, electropolymerization of monomer and reduction of electrolyte take place simultaneously to form a boron rich SEI layer, with electropolymerization being the dominant reaction along with ordinary SEI layer formation due to decomposition of electrolyte. Since electropolymerization is supposed to be the major reaction, this would result in homogeneous boron-rich interfacial layer that is expected to be highly stable and conductive in nature.

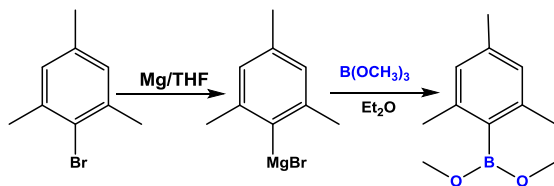
Experimental

Materials and instruments.—Mesitylbromide, 2-bromothiophene and thiophene (Thp) were obtained from TCI chemicals and were used without any further purification. Trimethoxyborane ($B(OCH_3)_3$), n-hexane and dehydrated methanol (MeOH) were obtained from Wako chemicals and were used as received. Mg turnings were purchased from Wako Chemicals and were activated by heating in vacuum prior to the reaction. Dehydrated THF and dehydrated ether were also bought from Wako chemicals and used as received for all the syntheses. Acetonitrile (Wako chemicals) was dried over activated molecular sieves (Wako chemicals) 5A 1/16 prior to use. AMImTFSI/BMImTFSI were synthesized after the ion exchange reaction of AMImCl (1-allyl-3-methylimidazolium chloride)/BMImCl (1-butyl-3-methylimidazolium chloride) with lithium bis-(trifluoromethanesulfonyl) imide (LiTFSI), purchased from

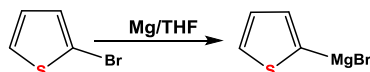
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Scheme 1. Synthesis of precursor 2,4,6-trimethylphenyldimethoxyborane (also referred as dimethoxy(mesityl)borane).



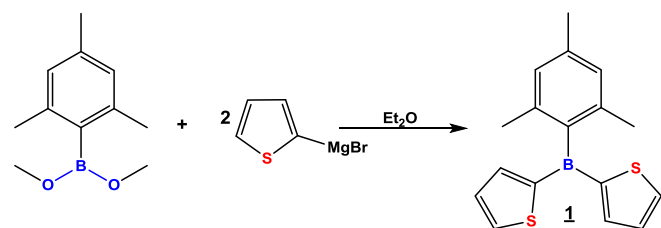
Scheme 2. Synthesis of thiophen-2-ylmagnesium bromide (Grignard reagent of thiophene).

Kanto chemicals. EC:DEC = 1:1 (v/v) solution was obtained from chameleon reagent and was used after drying over activated molecular sieves 5A 1/16.

NMR spectroscopic analysis was done by using Bruker model Avance III 400. The morphology of the electropolymerized electrodes were analyzed by using Scanning Electron Microscope (Hitachi H-4500). The prepared graphite based anodic half-cells were charged and discharged in a galvanostatic mode using compact charge and discharge system of EC Frontier; ECAD-1000. For charge-discharge studies, anodic half-cells were fabricated using CR2025 type coin cells. The cells were assembled using graphite/surface modified graphite as the working electrode (diameter = 15Φ), lithium metal as the counter electrode (15 mm, Honjo metals, Japan). The two electrodes were separated by a Celgard based separator cut in to a disk (diameter = 16 mm). EIS measurements were obtained before and after charge/discharge analysis using Biologic potentiostat. The impedance spectra were recorded under the frequency range from 0.01 Hz to 1 MHz using an AC amplitude of 10 mV at room temperature. Further, the spectra were fitted using ZSimpWin software from Princeton Applied Research.

Synthesis of boron-thiophene monomer.—Mesityldimethoxyborane was synthesized by a reaction of trimethoxyborane in dehydrated diethylether with mesitylmagnesium bromide (Scheme 1). Thien-2-yl magnesium bromide was prepared in a similar manner as reaction with mesitylbromide (Scheme 2). The final product, mesityldi(thien-2-yl)borane/boron-thiophene monomer (**1**) was synthesized by treating mesityldimethoxyborane (1.08 g, 0.0056 mol) with thien-2-yl magnesium bromide (2.10 g, 0.0112 mol) in dehydrated diethyl ether (Scheme 3). All the reactions were carried out under nitrogen atmosphere. Final product was first filtered, washed with hexane and recrystallized in methanol. The product was obtained in 95% yield (3.15 g) as off-white color crystals. The product was characterized using NMR and IR spectroscopy.

The synthesized low molecular weight boron monomer, **1** (B-Thp) was characterized by ^{11}B -NMR in CDCl_3 at 128 MHz showing a peak at 55.0 ppm (Fig. 1) referring to the incorporation of boron as an arylborane. Further, ^1H -NMR in CDCl_3 also supported the



Scheme 3. Synthetic scheme for mesityldi(thiophen-2-yl)borane, **1** (boron-thiophene monomer).

expected structure. ^{11}B -NMR (δ ppm, 128 MHz, CDCl_3 , Standard: Trimethoxyborane): 55.7 ppm; ^1H -NMR (δ ppm, 400 MHz, CDCl_3): 7.95 (s, Ar-CH (Thiophene), 2H), 7.82 (s, Ar-CH (Thiophene), 2H), 7.28 (s, Ar-CH (Thiophene), 2H), 6.88 (s, Ar-CH (mesityl), 2H), 2.38 (s, Ar-CH₃ (mesityl), 3H), 2.05 (s, Ar-CH₃ (mesityl), 6H).

IR spectrum of the compound, **1**, showed typical peaks due to the presence of thiophene and boron in the final product (Fig. 2). The observed IR frequencies are listed as: 3000–2950 cm^{-1} , aromatic CH stretching conjugated with C = C (Thiophene and mesityl ring); 1550–1195 cm^{-1} , C = C in-plane vibrations (thiophene); 1020 cm^{-1} , B-C bond stretching; 600–500 cm^{-1} , mesityl ring out of plane deformation vibrations.

Electropolymerization methods.—The synthesized monomer was electropolymerized in two different ways:

1. **Pre-modified Anodes:** The monomer was electropolymerized by carrying out cyclic sweeps between -2.0 to 2.0 V at a scan rate of 0.05 V/s using Ag/Ag^+ as the reference electrode in 0.02 M **1** in ionic liquid (AMImTFSI/BMImTFSI): acetonitrile = 7:3 (v/v) system as electrolyte. The working and the counter electrodes were graphite and Pt chip respectively. A box cell (flat-cell) was used in this method of electropolymerization with N_2 purging. The electropolymerized samples were studied visually and by using SEM. The SEI characteristics were studied by using these pre-modified electrodes for charge-discharge analysis.
2. **In-situ modified Anodes:** The second method was electropolymerization while the electrode was subjected to charge-discharge. Graphitic anode half-cells were fabricated using Li foil as the reference and counter electrode with 0.1 M **1** in 1 M LiTFSI in EC:DEC = 1:1 as the electrolyte. The separator was dipped into the electrolyte and was used in the cell with additional 30 μL of the electrolyte. Repeated charge-discharge cycles while monitoring the overall impedance changes before and after the charge discharge led to the understanding that the electropolymerization of the monomer **1** has been accomplished. This data was compared with an anodic half-cell with 0.1 M thiophene in 1.0 M LiTFSI in EC:DEC = 1:1 as the electrolyte.

In both the cases, a very dilute monomer concentration of the conducting monomer was taken (0.01 M thiophene/**1**).

The electropolymerized pre-modified electrodes were morphologically characterized by SEM which showed the formation of the smooth and compact layer on the graphite electrodes. A thick polymerized coating was formed which was indicated by the formation of a blue-green layer onto the carbon. Fig. 3 shows the image of the carbon anode after electropolymerization and the SEM image exhibiting the layer-by-layer polymerization of monomer **1** on carbon anode.

Results and Discussion

Pre-modified anodes.—The formation of electropolymerized layer onto the graphite anodes can be clearly seen in SEM images and further, was supported by electrochemical impedance spectroscopic analysis. Impedance spectroscopy measurement was carried out after every 5 cycles within the frequency range of 1 MHz to 0.1 Hz.

Till 15 cycles there was only one semicircle observed. Appearance of a second distinct semi-circle or a time constant after 15 cycles referred to the second phase can be associated with the formation of a polymerized layer. The impedance spectra for all the sets of 5 cycles are shown in Fig. 4. In the case of electropolymerization, where formation of a conductive surface occurs on an electrode, the migrating charged particles are associated with surface diffusion-related relaxation processes. These charged particles also have a tendency to agglomerate and form a capacitive or resistive conduction path. Therefore, the formation of continuous clusters of conductive particles can be explained by a single time-constant/semi-circle in Nyquist complex impedance plot. Such an impedance spectrum is modeled by a two-component R_1/C_1 - R_2/C_2 model and the electrical properties

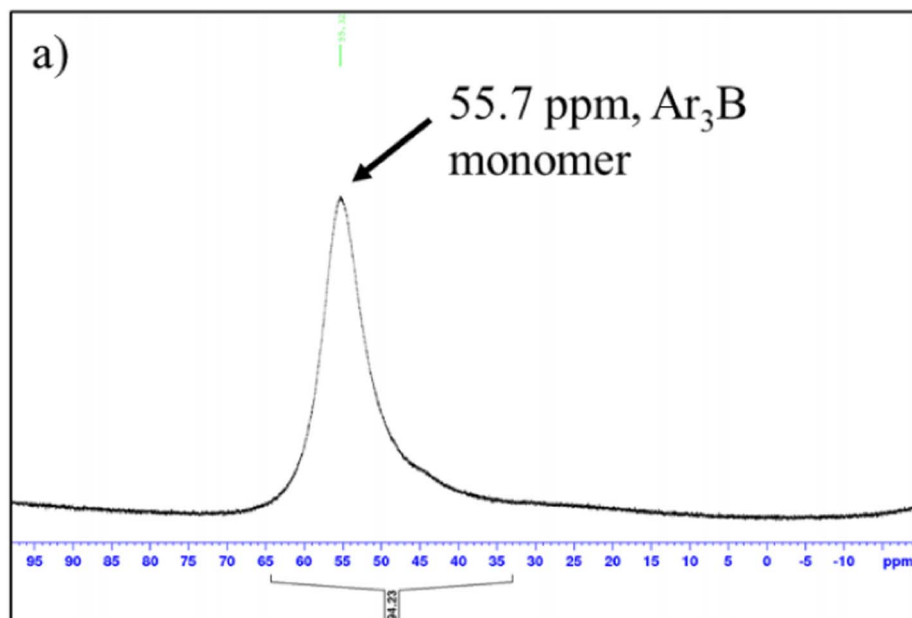


Figure 1. a) ^{11}B -NMR and b) ^1H -NMR spectra of **1** (structure given in Fig. 1b) in CDCl_3 .

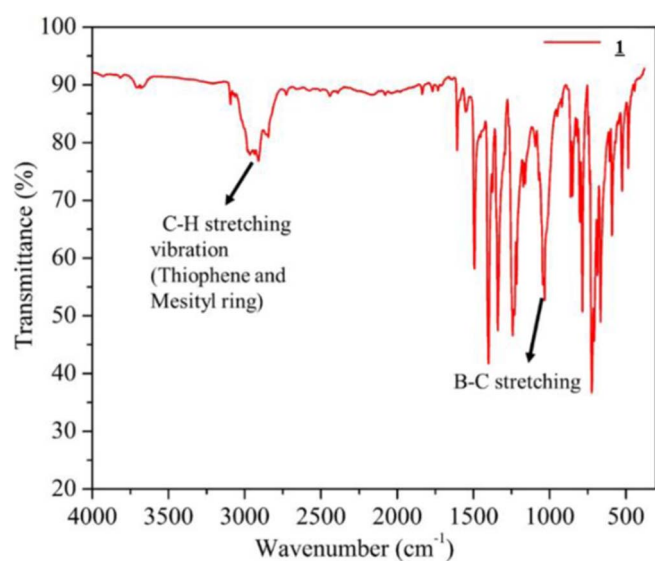
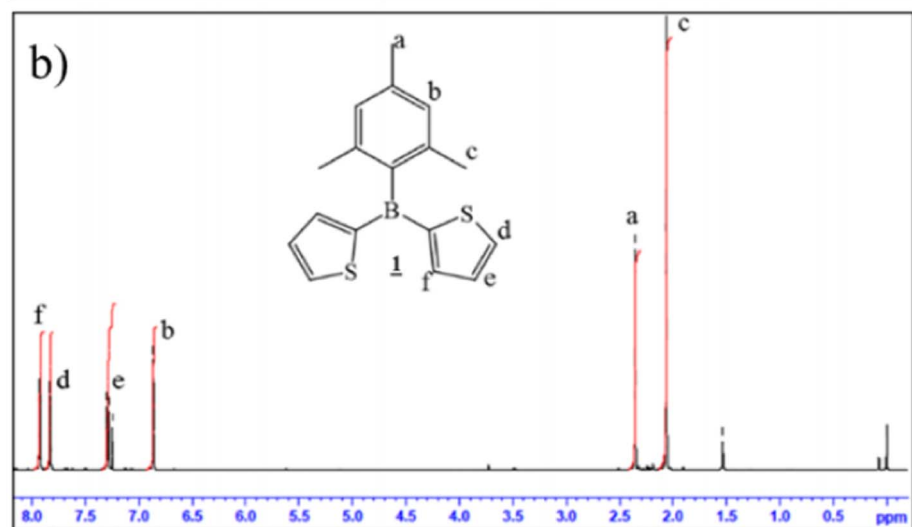


Figure 2. IR spectrum of **1**.

of such a system are demonstrated as a function of concentration or fraction of conductive particle, given by “n”.²⁷

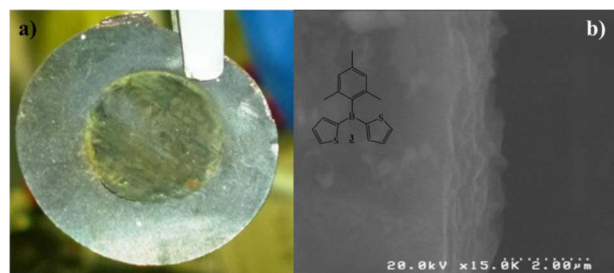


Figure 3. a) Image of carbon electrode after electropolymerization and b) SEM image showing layer-by-layer formation of polymer during the repeated cycles.

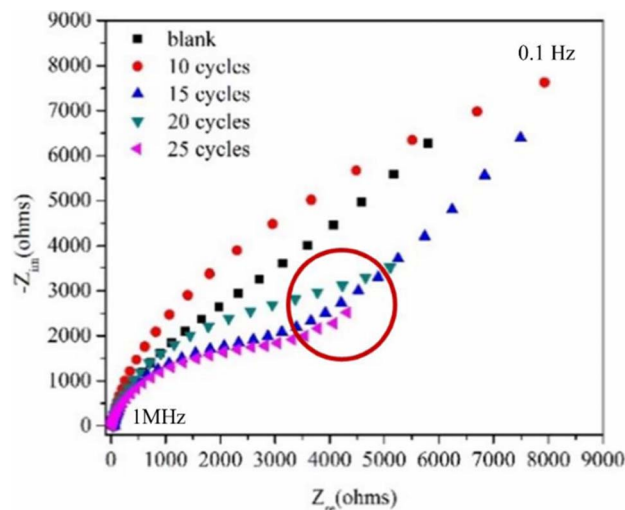


Figure 4. Impedance spectra taken in the frequency range 1 MHz to 0.1 Hz of electropolymerization with each set of 5 cyclic voltammetric cycles. The circle portion (marked with red) shows the formation of a second semicircle.

R_1 and C_1 being the bulk/solution resistance, and
 R_2 and C_2 being capacitance and resistance of the individual conducting particles responsible for the cluster formation.

This spectral analysis by the electric circuit model results in:

- A value of $n < 0.25$, implies negligible agglomeration of conductive particles into organized orientated clusters, and hence only a single-time constant appears.
- For $n > 0.5$, the electrodes become shortened due to over agglomeration or development of continuous cluster of conductive particles and hence, spectrum is resolved as a single time constant.
- For value of $0.25 < n < 0.5$, i.e. intermediate concentration of the conductive medium and random distribution and agglomeration of conductive particles, an additional time constant emerges in medium frequencies (which is also seen in Fig. 4).

The explanation above justifies the appearance of one semi-circle for low concentration of the conductive particle (i.e. when the electropolymerization has initiated) and the appearance of second semi-circle after 15 cycles which is related to the formation of an electropolymerized layer.

The pre-modified anodes were tested for charge-discharge behavior in an anodic half-cell. The charge-discharge behavior of pre-modified anode was compared with bare graphite for 10 continuous charge-discharge cycles at a rate of 0.5 C. Both the electrodes showed typical charge-discharge profiles showing intercalation at lower potentials. The uncoated or the bare graphite electrode showed a drastic decrease in the capacity with each cycle (Fig. 5a). This decrease in capacity from 200 mAh/g (1st cycle) to approx. 30 mAh/g (10th cycle) can be due to the formation of a non-compatible SEI layer which is insulating in nature. This was also asserted by the impedance spectra

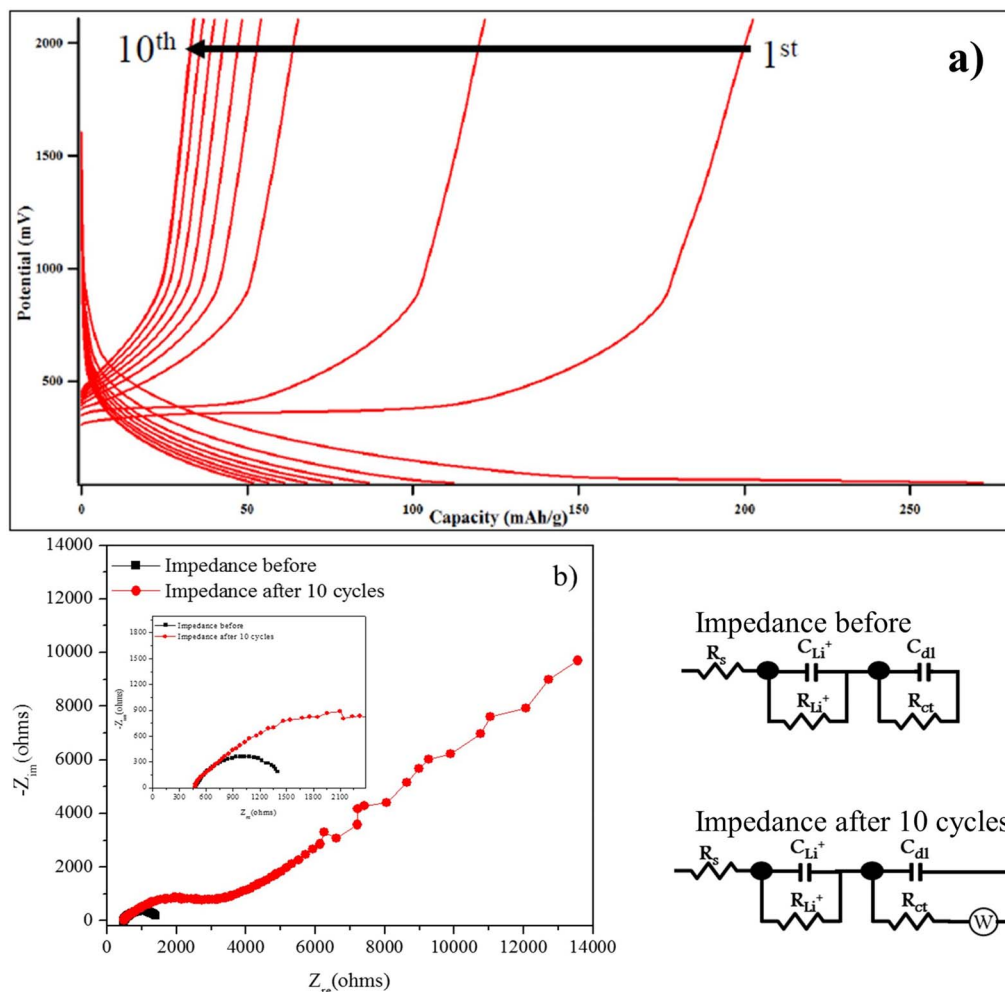


Figure 5. a) Charge discharge profiles of anodic half-cell using bare graphite with 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte at 0.1 C and b) Impedance analysis of anodic half-cell using graphite with 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte at 0.1 C.

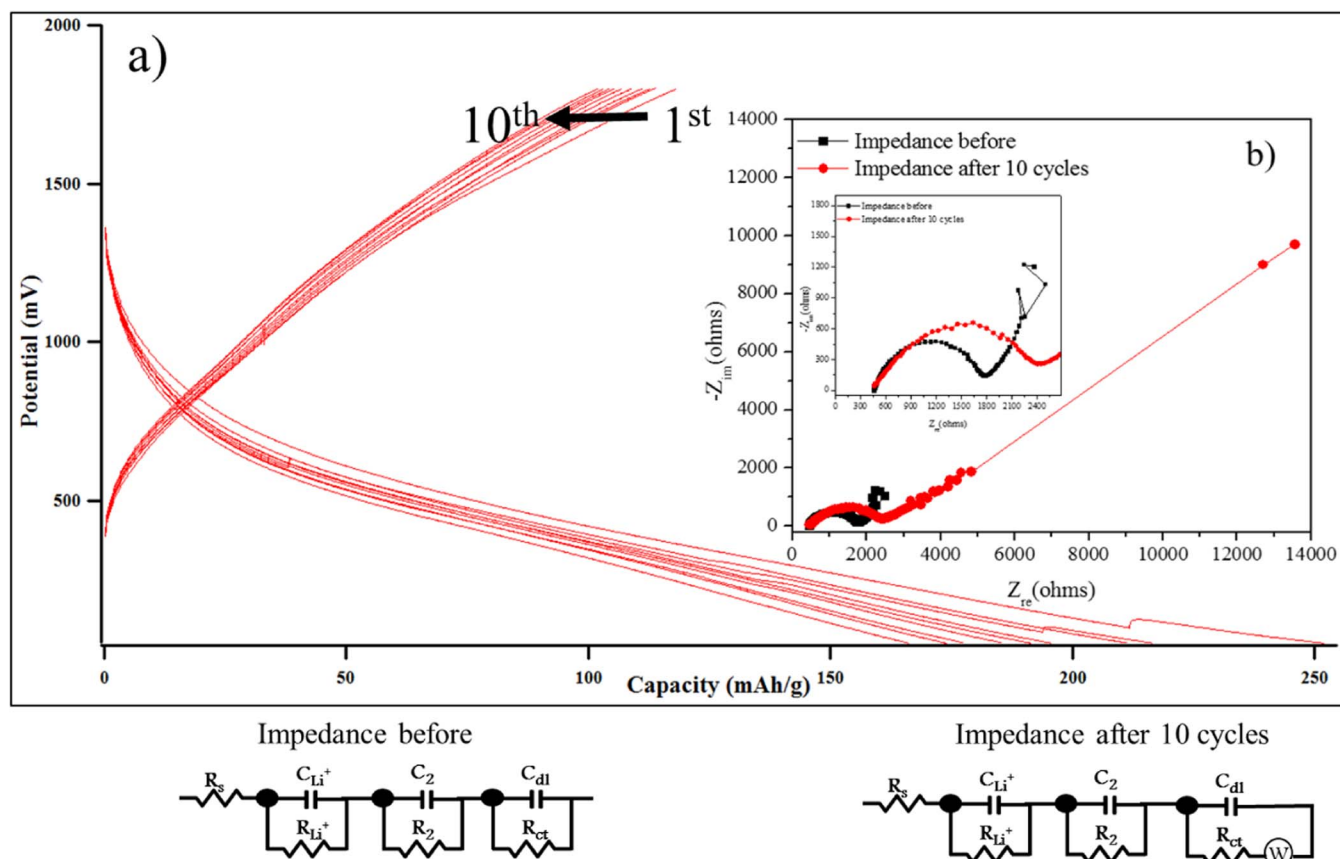


Figure 6. a) Charge-discharge profiles of anodic half-cell using electropolymerized graphite with monomer **1** and 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte at 0.1 C and b) Impedance analysis of anodic half-cell using electropolymerized graphite with monomer **1** and 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte at 0.1 C.

as shown in Fig. 5b depicting an increase in the interfacial resistance from 912.6 Ω to 3133 Ω after 10 cycles.

In contrast, the charge-discharge cycles of graphite electrodes on which electropolymerization of boron-thiophene monomer was carried out did not show this capacity fading effect. In the case of boron-thiophene coated electrodes, the capacity for the 10 cycle was maintained at 125 mAh/g (Fig. 6a) and the results from impedance analysis also correlated well by showing marginal increase from 583.1 Ω to 2024 Ω (Fig. 6b) unlike the one with bare graphite which showed 3133 Ω . This can be attributed to the polymerized boron-thiophene film which might be acting as a conducting SEI layer and have prevented decomposition of any electrolyte over the graphite.

In-situ modified anodes.—A method was next examined to conduct the electropolymerization of thiophene derivative as the major dominant reaction during the SEI formation with simultaneous reduction of electrolyte. Hence, in-situ modification of graphite electrode in galvanostatic mode was carried out.

In general, oxidation of thiophene when carried out via recurring cyclic potential sweeps occurs at 1.4 V–2.3 V vs SCE (Saturated calomel electrode). Further, in acetonitrile-TBAP (tetrabutylammoniumperchlorate) systems, electropolymerization occurs at 1.6 V vs SCE. However, it is known that in galvanostatic mode, electropolymerization occurs at 0.5 V more positive than the oxidation potential of the monomer and results in more conducting and homogeneous polymerized films.²¹ In this case, the oxidation of boron-thiophene monomer in an anaerobic pressurized CR 2025 cell condition, occurs at 0.6 V vs. Li/Li⁺ when performed via cyclic sweep method and at 1.1 V in galvanostatic mode.

Anodic half-cell with bare graphite was repeatedly cycled at 0.1 C with 0.1 M LiTFSI in EC:DC = 1:1 as the electrolyte. Fig. 7 shows the charge-discharge profiles and impedance spectra using thiophene as the additive. In this case, we observed an increase in the discharge capacity from 1st (21 mAh/g) to 7th cycle (58 mAh/g) and then, the capacity decreased from 7th to 10th cycle (50 mAh/g). The impedance analysis showed a drastic decrease in the interfacial resistance from 469.9 Ω (before charge-discharge) to 55.86 Ω after 10 cycles. This was due to the use of thiophene as additive for modifying the electrode surface which made it electronically conductive. A potential plateau was observed at 1.6 V in the charge-discharge curve corresponding to the electropolymerization of thiophene. Next, surface modification was done using borylated-thiophene, **1** as the additive with 0.1 M LiTFSI in EC:DC = 1:1 as the electrolyte. As compared to thiophene, an improvement was seen in the charge-discharge cycles as well as in the reduction of interfacial resistance as seen in Fig. 8.

The discharge capacity for the first cycle was observed to be 60 mAh/g and the rest of the cycles showed an average discharge capacity of 80 mAh/g (Fig. 8) which was higher as compared to the one with thiophene as the additive. A plateau of slight dip at 1.1 V in the charging cycle referred to the electropolymerization of the boron-thiophene monomer. The charge transfer resistance before charge-discharge was 2311 Ω . However, a drastic fall was seen in the charge-transfer resistance (10.98 Ω) after in-situ electropolymerization of boron-thiophene. This decrease was due to the result of incorporation of boron originating from the monomer. As boron is electron deficient, its presence at the interface will result in increased interaction with the Lewis base electrolyte and reduction of the electrode-electrolyte interfacial resistance. This polymerized layer of **1** formed onto the

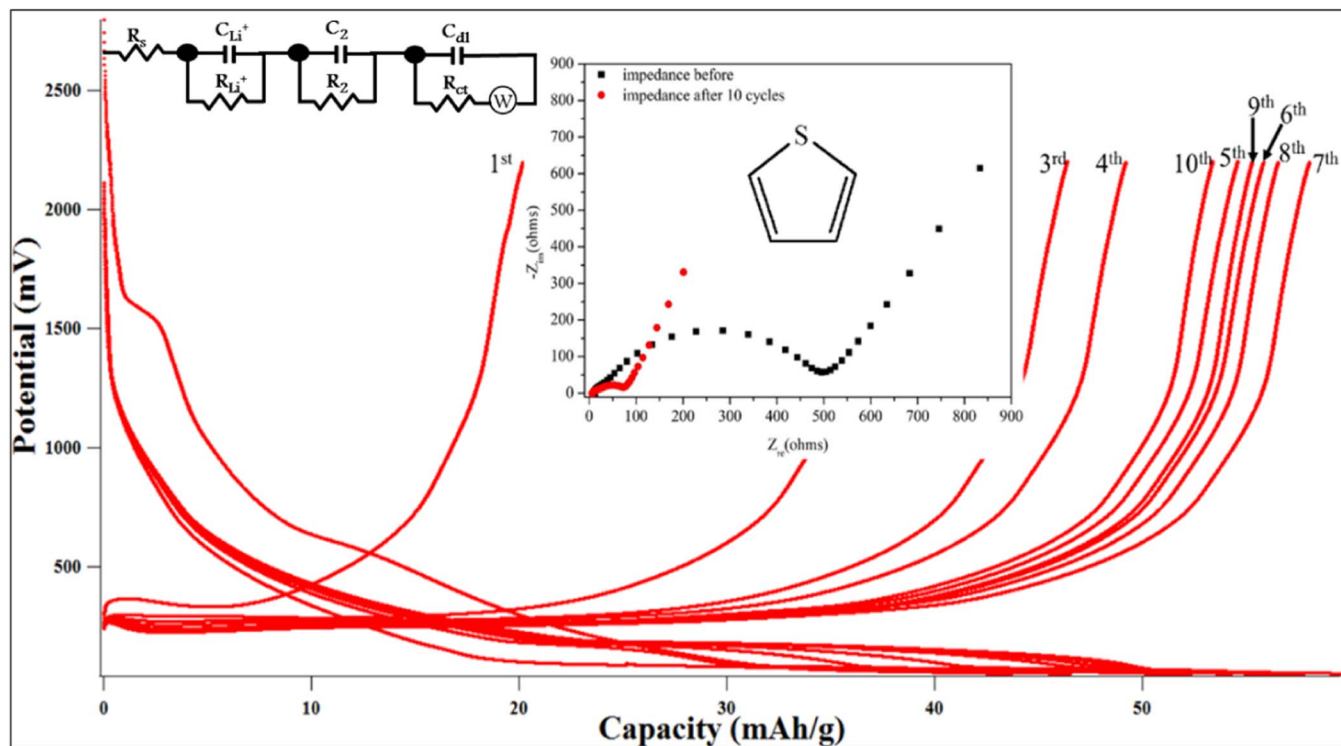


Figure 7. In-situ electropolymerization via repeated charge discharge cycles and impedance analysis of anodic half-cell using 0.01 M thiophene in 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte under CC mode at 0.1 C.

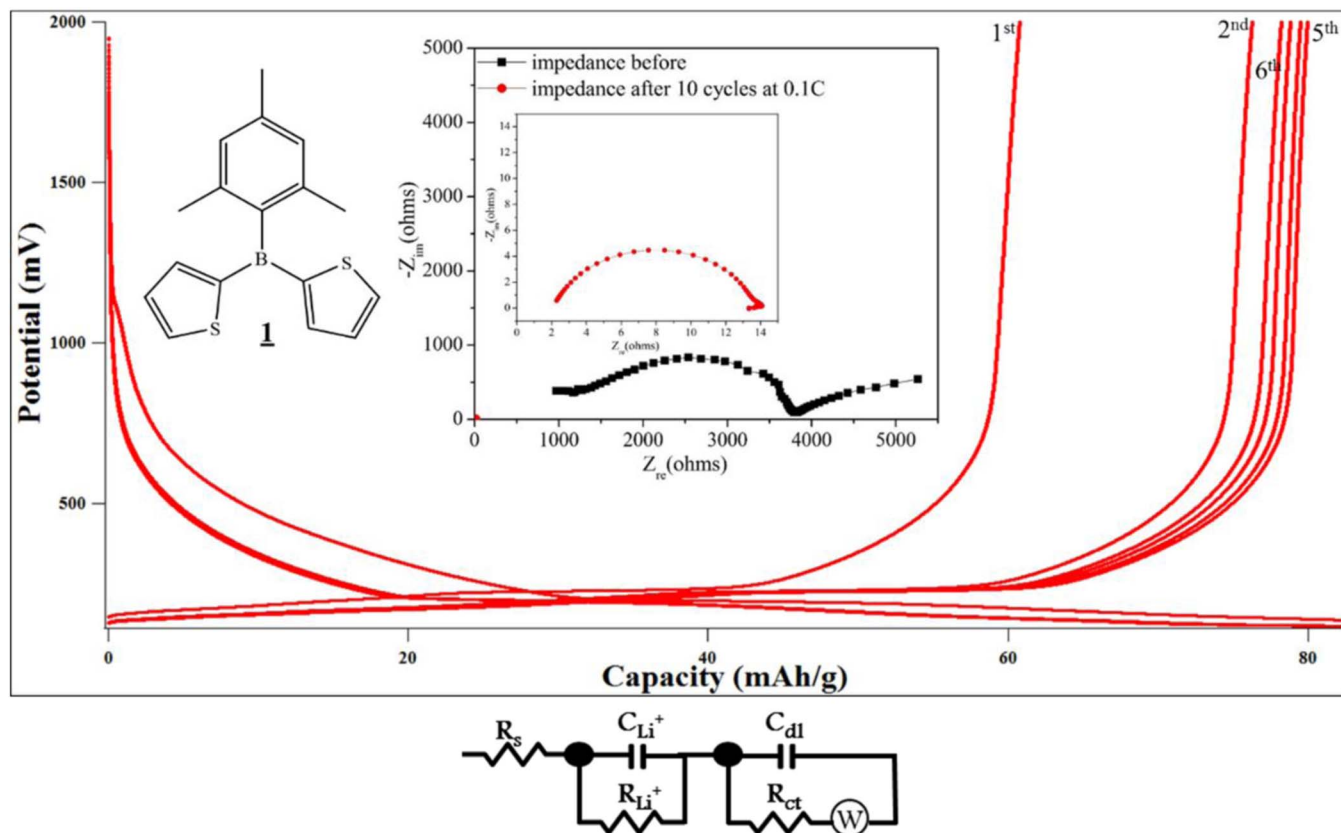


Figure 8. In-situ electropolymerization via repeated charge discharge cycles and impedance analysis of anodic half-cell using 0.01 M **1** in 0.1 M LiTFSI in EC:DC = 1:1 as electrolyte under CC mode at 0.1 C.

Table I. Summary of results of charge-discharge and impedance measurements.

Graphite with additive	Surface modification of anode			In-situ electropolymerization		
	Capacity from 1 st -10 th cycle (mAh/g)	Cyclability	Interfacial Resistance after cycling (Ω)	Capacity of 1 st cycle (mAh/g)	Cyclability	Interfacial Resistance after cycling (Ω)
No additive	200 to 30	Poor	3133	-	-	-
Thiophene	-	-	-	21	Poor	55.86
1	125-100	Good	2024	60	Good	10.98

graphite electrode can be hypothesized to form simultaneously with the formation of electrolyte derived SEI layer leading to a large decrease in the interfacial resistance. Table I gives a comparison of charge-discharge capacities and interfacial resistances of pre-modified anodes and in-situ modified anodes.

The method of surface modification significantly influence the equivalent circuits used for fitting the impedance spectra. As shown as inset in each of the charge-discharge profile (Figs. 5–8), the equivalent circuit varies with both the method of electropolymerization as well as the monomer used. Observing closely, it can be noted that, two types of electrical circuits are used, *a*) 3-element circuit and *b*) 4-element circuit. Element here refers to a single resistor (R), a capacitor (C) or a parallel combination of the constant phase element (Q) with resistor and Warburg element (W) in series (Q(RW)). In the case of graphite (Fig. 5), since there is no monomer present, a 3-element circuit satisfies the spectral fitting defining the specific electrochemical processes, (i) Li migration through the solution at high frequencies giving the solution resistance, (ii) Li migration within the electrode surface coupled with the double layer capacitance and (iii) Charge transfer process between the electrode and the electrolyte.⁶ However, in the case of electrodes pre-modified with incorporation of monomer **1** (Fig. 6) by electropolymerization (via CV) followed by charge-discharge (via galvanostatic cycling), SEI formation occurs as two different processes. The formation of the solid electrolyte interface (SEI) layer occurs over the already thick electropolymerized layer, i.e., the active material is separated by the thick electropolymerized layer from the SEI layer. Hence, a 4-element circuit satisfies the spectra. Whereas, for in-situ modified anodes, it can be hypothesized that in case of thiophene as the monomer for in-situ modification (Fig. 7), a 4-element circuit suits perfectly as the formation of SEI layer and the electropolymerized layer occurs as two different layers. However, in the case of in-situ modification using boron-thiophene monomer **1**, the formation of electropolymerized layer happens in integration with the solvent derived SEI layer. Since presence of boron tends to improve the contact at electrode-electrolyte interface, the SEI layer formation while galvanostatic cycling interpenetrates the electropolymerized layer making it a more conductive single compact layer. Due to this, the cell impedance appears to be low with increasing capacity. All the results can be corroborated best based on the three element circuit. Tables II and III list all the electric circuit fitting results in a tabular format. (Fig. 8). Fig. 9 illustrates the formation of

Table III. Impedance fitting results for In-situ-modified electrodes.

Imp	In-situ			
	Thiophene		Boron-Thiophene	
	Before	After	Before	After
R1	10.52	6.43	906.6	2.574
R2	18.31	13.69	511.9	104.4
R3	469.9	55.86	2311	10.98
R4	6994	22.63		
Chi-square	2.0e-3	2.4e-4	4.9e-3	6.7e-3
Circuit	R(QR)(QR) (Q(RW))		R(QR)(Q(RW))	

SEI layer and electropolymerized layer over the graphite anode via the two modes of electropolymerization.

Conclusions

A low molecular weight monomer, termed as borylated-thiophene monomer **1** was synthesized and was electropolymerized over carbon anodes to study the change in interfacial resistance. **1** was used for surface modification of anodes (graphite) via cyclic voltammetry in the presence of ionic liquids. Surface modification was done in two ways: 1. by pre-modifying the anodes and then using them for battery testing after characterization by SEM; and 2. by in-situ modification of bare graphite anodes for charge discharge cycles in the presence of boron-thiophene monomer as additive in the conventional electrolyte. The results obtained with the monomer were compared with the bare graphite electrodes. It was observed that in the case of pre-modified anodes, in the presence of boron-polymer membrane, the cyclability was improved, however, the charge-transfer resistance increased with a smaller value. Increment of charge transfer resistance can be due to too large thickness of SEI layer. In case of in-situ modified anodes, the capacity obtained using boron-thiophene monomer as additive was higher as compared to thiophene. A drastic fall in charge-transfer resistance in the lithium cell containing the boron-thiophene monomer asserted the charge-discharge data without any fading and can be

Table II. Impedance fitting results for Pre-modified electrodes.

Imp	Pre-modified			
	Bare Graphite		Boro-thiophene	
	Before	After	Before	After
R1	476.9	452.1	468.1	755.2
R2	912.6	3133	3460	398.6
R3	109.3	646.1	451.6	1012
R4			1315	1112
Chi-square	2.28 e-5	3.4e-4	1.2e-4	1.1e-4
Circuit	R(QR)(QR)		R(QR)(QR)(QR)	

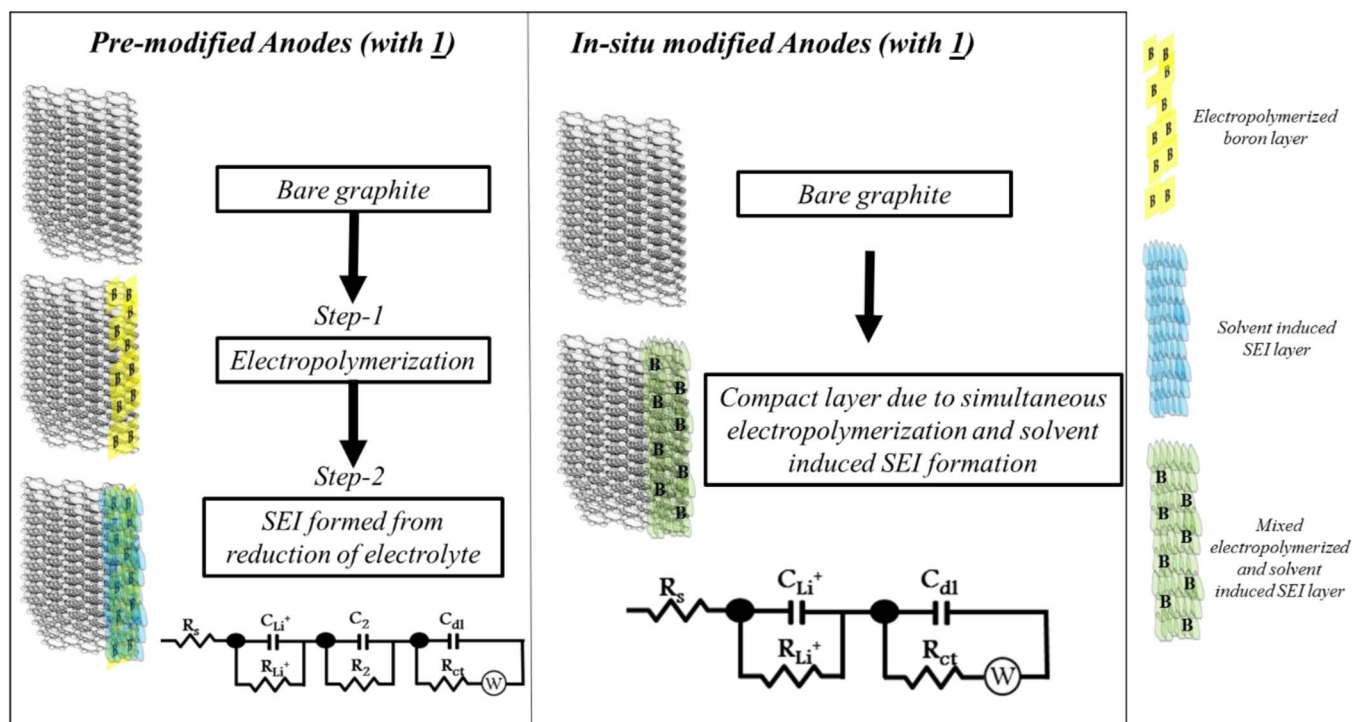


Figure 9. Formation of electropolymerized layer and solvent induced SEI layer over graphite anode via the two methods of surface modification.

attributed to the formation of a highly stable and boron-rich conductive interface.

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